

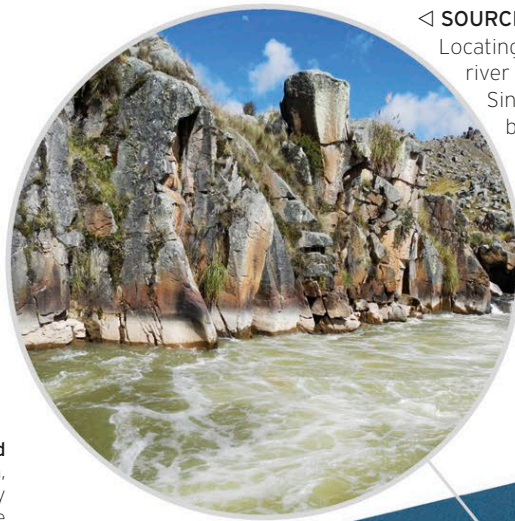
The features of a RIVER BASIN

A river basin is a geographic area in which all of the water that enters, usually by precipitation, drains to a common outlet through a river and its network of tributaries. River basins vary greatly in shape, complexity, and size—with the Amazon Basin being by far the largest on Earth. However, all river basins share a set of common features.

Source to mouth

A basin's watershed marks its perimeter, separating it from the basins of adjacent river systems. Naturally, watersheds run through areas of high ground, such as mountain ranges and plateaus. The source of a river system—which may be a spring, glacial meltwater, or a marsh—is the farthest point of a river's course from its eventual outlet, measured along the river's course itself. The elevated sections of a river basin feature streams that are usually fast flowing, with rapids and waterfalls, that descend onto lower, flatter terrain and become slower and wider, and often meander.

Most channels in a basin are tributaries that empty into the main river. Tributaries, such as the Amazon's Rio Negro, may themselves be substantial rivers, and waters from disparate sources, of different content and color, may come together at their meeting points, or confluences. A mature river in its lower course usually creates a floodplain by dropping sediment on its valley floor during floods. It meanders across this plain, its course shifting, either gradually as it erodes its banks or abruptly after severe floods. Most river basins—but not all (see p.197)—empty into an ocean or a lake. A river's mouth usually features an estuary where freshwater and saltwater mix, and often a delta, formed by the accumulation of sediments carried by the river.



◀ **SOURCE**
Locating the source of a great river is often contentious. Since 2014, scientists have believed the Amazon's source to be in the Mantaro river basin, high in the Peruvian Andes.

Watershed
The boundary of a river basin, called the watershed, commonly runs along a mountain ridge. The Andes form most of the western watershed of the Amazon.

Pacific Ocean

Upper course
In the upper course of a river, it tends to descend steeply and is joined by many small tributaries. In the Amazon's upper course, rivers and streams cascade down steep slopes of the Andes.

region of seasonal flooding

▶ **CATCHMENT AREA**
A river basin, also called its catchment area, encompasses the steep upper courses of a river system as well as the gentle gradients of the river's lower courses. In the Amazon's lower course, the river flows over such flat terrain that it falls only 230 ft (70 m) from the city of Manaus to the sea—a distance of some 820 miles (1,400 km).



△ **AQUATIC ACROBAT**
Unlike oceanic dolphins, the Amazon river dolphin has a flexible neck, which helps it negotiate vegetation in the flooded forest. The dolphins are seen lifting their heads clear of the water to check the surroundings, and occasionally they jump up to 3 ft (1m) above the water's surface.

REGIONAL FLOODING

Every year, vast areas of the Amazon's floodplains become inundated with water, creating the most extensive system of river-flooded forests in the world. The weight of water is so great that GPS sensors show it depressing the land by 3 in (7.5 cm) during peak flood. Due to the Amazon Basin's vast size, and the uneven distribution of its seasonal rainfall, different regions in the basin are affected at different times.

